



Bisphenol F induces nonalcoholic fatty liver disease-like changes: Involvement of lysosome disorder in lipid droplet deposition[☆]

Jun Wang^{a, c}, Pengfei Yu^a, Xuexue Xie^a, Linlin Wu^e, Manfei Zhou^b, Fei Huan^a, Lei Jiang^d, Rong Gao^{b, *}

^a Key Lab of Modern Toxicology (NJMU), Ministry of Education; Department of Toxicology, School of Public Health, Nanjing Medical University, 818 Tianyuan East Road, Nanjing, Jiangsu, 211166, China

^b Department of Hygienic Analysis and Detection, Key Laboratory of Modern Toxicology, Ministry of Education, School of Public Health, Nanjing Medical University, Nanjing, China

^c China International Cooperation Center for Environment and Human Health, Nanjing Medical University, 818 Tianyuan East Road, Nanjing, Jiangsu, 211166, China

^d Department of Emergency Medicine, The First Affiliated Hospital of Nanjing Medical University, 300 Guangzhou Road, Nanjing, Jiangsu, 210029, China

^e Wuxi Center for Disease Control and Prevention, Wuxi, Jiangsu, 214000, China

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ABSTRACT

Epidemiological studies have demonstrated that the general population's exposure to bisphenol A (BPA) substitutes is ubiquitous. Bisphenol F (BPF), one of the main BPA substitutes, is increasingly replacing BPA in plastics for food and beverage applications. Accumulating evidence suggests that BPA exposure is associated with nonalcoholic fatty liver disease (NAFLD)-like changes. However, the potential effects of BPF on lipid homeostasis remain poorly understood. In the present study, an epidemiological analysis with LC-MS-MS revealed that the BPF concentrations in the serum of NAFLD patients were significantly higher than those in a control group. Supporting this result, using Oil Red O, BODIPY 493/503, LipidTox Deep Red staining and gas chromatography-time-of-flight mass spectrometry (TOF-MS) assays, we found that BPF exposure induced NAFLD-like changes, with obvious lipid droplet deposition, triglyceride (TG) and fatty acids increase in mouse livers. Meanwhile, lipid droplet deposition and TG increase induced by BPF were also observed in HepG2 cells, accompanied by autophagic flux blockade, including autophagosome accumulation and the decreased degradation of SQSTM1/p62. Using adenoviruses dual-reporter plasmid RFP-GFP-LC3, RFP-GFP-PLIN2 transfection, AO staining, and EGFR degradation assays, we demonstrated that BPF treatment impaired lysosomal degradative capacity, since BPF treatment obviously impaired lysosomal acidification, manifested as decreased lysosomal hydrolase cathepsin L (CTSL) and mature cathepsin D (CTSD) in HepG2 and mouse liver issues. Additionally, v-ATPase D, a multi-subunit enzyme that mediates acidification of eukaryotic intracellular organelles, significantly decreased after BPF exposure in both the vitro and in vivo studies.

This study ascertained a novel mechanism involving dysfunctional of lysosomal degradative capacity induced by BPF, which contributes to lipophagic disorders and causes lipid droplet deposition. This work provides evidence that lysosomes may be a target organelle where BPF exerts its potential toxicity; therefore, novel intervention strategies targeting lysosome are promising for BPF-induced NAFLD-like changes.

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* Corresponding author. Department of Hygienic Analysis and Detection, School of Public Health, Nanjing Medical University, 818 Tianyuan East Road, Nanjing, Jiangsu, 211166, China.

E-mail address: gaorong@njmu.edu.cn (R. Gao).

1. Introduction

Since 1983, the prevalence of nonalcoholic fatty liver disease (NAFLD) has increased more than 20 times worldwide. Over the past decade, the prevalence of NAFLD in Western countries has risen to 20–30% in the general adult population. In diabetic and obese patients, the prevalence of NAFLD is as high as 60–70%